

Don't-care minimization

Some input combinations never happen, or the output does not matter when they do

ON, OFF, and optional

- Every product term must cover all ON minterms
- Cubes must never include an OFF row
- Don't-care rows are optional, you choose zero or one to enlarge groups
- Sigma-m with d lists minterms and don't-cares
- Compare covers side by side to see the shrink

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Don't-care minimization

Mark truth-table rows as **0**, **1**, or **X**. X entries need not be covered — using them as 1s often yields a smaller SOP.

Starter example: 3-var $\Sigma m(0, 2, 5, 7) + d(1, 3)$ — without X: $A'C' + AC$; with X: $A' + C$.

Load starter example

Challenges (1/22)

Quiz: X. Don't-care rows are written as? Answer: X

Your answer

Show hintCheckNextIdle

quiz-xquiz-mustquiz-optionalquiz-offstarter-loadstarter-withoutstarter-withminimize-clickcompare-clickquiz-shrinktwo-vartwo-withoutclear-xtoggle-dcquiz-sigmaexplainquiz-winand-no-helpstarter-shrunk-flagthree-offquiz-qmfull-compare

Core ideas

ON-set
Rows $F=1$ — every cover must hit these.

Don't-care
Rows $F=X$ — free; may join cubes as 1s.

OFF-set
Rows $F=0$ — cubes must never include these.

Don't-care lab starter

learn_digital — dont care lab

Workbook practice

- In the workbook track, take the starter pattern by hand
- Write the ignore-X SOP, then regroup using X at one and three
- Confirm the with-X form has fewer terms
- Try a two-variable preset with one ON and one X
- Name one pitfall: treating X as OFF by default and missing a smaller cover

Pitfalls to watch

- Do not let a group cover an OFF row
- X is not “don’t optimize”, it is “pick whichever helps.” And remember
- Real designs still need a spec for what is truly unreachable versus merely unused

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, minimize one small table with at least one X
- When you are ready, take the short quiz, then continue to logic hazards